

University of Houston – Spring 2024

STRATEGIC SOURCING CASE STUDY

On Monday morning Chris Jackson, Global Sourcing Specialist for subsea trees, was beginning to plan out his week when Lori Grant, Global Planning Manager, walked into his office to deliver the details of a new project. TechnipFMC has recently received a contract from a major operator to provide 40 next generation (Subsea 2.0) Subsea Christmas Trees for a new offshore field in the Gulf of Mexico. A Subsea Tree is a piece of equipment that is fixed to the wellhead of a completed well for the purpose of monitoring and controlling the flow of hydrocarbons. As there are several wells drilled for a project, each wellhead will need a Subsea Tree to regulate and maximize production.

As a sourcing specialist in the tree group, Chris and his team have been tasked with developing a sourcing strategy for a portion of the Bill of Materials (BOM). A major factor influencing your strategy will be the recent decision to consolidate in-house manufacturing and assembly located in South America. Further, some of the items you are being asked to source can also be made in-house. As a results, you and your team will need to include a make/buy recommendation in your strategy.

As Chris begins to prepare for the project he takes a moment to reflect on the company's core values, behaviors and foundational beliefs:

 Realizing possibilities We strive for ever better We take initiative We learn from success and failure	 Achieving together We work as one team We share knowledge We embrace diversity of thought	 Building trust We listen to improve We partner constructively We seek to outperform
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In everything we do, we never compromise on:

Safety	Integrity	Quality	Respect	Sustainability
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Our foundational beliefs reflect how we fundamentally do business and are things we'll never compromise on.

- **Safety:** We will not compromise on health, safety and security
- **Integrity:** We hold ourselves to the highest moral and ethical principles
- **Quality:** We deliver the highest quality in everything we do
- **Respect:** We treat everyone honestly, fairly, and courteously
- **Sustainability:** We act responsibly, always considering our impact on the planet, people, and communities in which we operate.

Going Forward

After this initial meeting, Chris begins thinking about the information he and the team will need to gather and analyze to develop the sourcing strategy. In order to be successful, they will need to create a presentation that addresses the following areas:

1. Summary of Case Facts and Problem Solving Methodology
2. Executive Summary of Selected Strategy
3. Recommendations and Data Analysis
4. Implementation and Follow Up Plan

Knowing they will need to collect data for the project, Chris makes a list of people they will need to meet with over the coming days.

Department / Function	Reason for Meeting
Global Planning Manager	Project requirement dates, bill of materials (BOM), and internal manufacturing capabilities.
Category Management	As this project is for a next generation design, the category teams will need assess the capabilities required to produce the new parts. The Category Managers for each commodity will provided you with a list of qualified suppliers and up-to-date information on volume discounts.
Supplier Quality	Cost / benefit information related to supplier development.
Supply Chain Analyst	Supplier performance metrics and SPI.
Engineering	Issues related to engineering changes and late releases
Procurement	Buyers contact the qualified vendors and provide price and lead times for all parts.
Logistics	Logistics will provide you with the time and cost for transporting goods to each manufacturing plant.

Notes from Meeting with Global Planning Manager

- Planning has decided to assemble in TechnipFMC's South American Plant. Lori provided a table detailing the quantities and the planned finish dates.
- All parts must arrive one month ahead of the planned finish date to allow time for finally assembly of components.
- Bill of material represents the part requirements for a single subsea tree.
- Planning will load sales orders at the end of October 2024. As a result, requisitions will not be able to be converted into purchase orders until November 1st 2024.
- Parts with criticality ratings of 8 and above pose serious potential consequences and require an additional 30 days of lead time due to pre-production supplier documentation review.
- Lori also provided a list of in-house manufacturing capabilities and associated production costs for the two manufacturing locations.
 - a. Direct material and labor represent 60% of the total cost of parts manufactured in house.

- b. The new design for the **Composite Valve Block** represents a major engineering achievement for the company and provides a distinct competitive advantage over alternatives in the market.
- c. Though TechnipFMC has the machining capabilities to make **tubing hangers**, reworks have become a regular occurrence. Over the past 6 months, TechnipFMC has had issues meeting customer requirements for this part.
- Assume **no inventory holding costs will be incurred** either in-house or outsourced parts.
- When making sourcing and investment decisions you and the team should keep in mind that **demand for subsea systems is expected to increase for the next 5 years.**

See Exhibit 1 - Schedule

Requirement Dates Tree Deliveries

Planned Assembly Finish Dates	South American Plant
Sept. 1 st 2025	10
Dec. 1 st 2025	30

Bill of Materials for Next Generation Tree

COMPONENT	CRITICALITY	QTY	CATEGORY
Composite Valve Block	10	1 ea	Machining
Gate Valve	7	2 ea	Machining
Metal Seal	8	5 ea	Machining
Tubing Hanger	10	1 ea	Machining
Wing Blocks	5	2 ea	Machining

See Exhibit 2 – Manufacturing Costs & Capacity

Notes from Meeting with Category Management

The management of TechnipFMC's supply base is critically important. With suppliers providing upwards of 90% of sub assembly components, TechnipFMC **must insure that QSL** (Qualified Suppliers List) **has the right capabilities and enough capacity to meet current and future demand.** Further, the SQDC (Safety, Quality, Delivery, and Cost) for all suppliers must be continuously monitored. Suppliers are organized by the service they provide and grouped into categories. **The category you are sourcing is Machining.**

See Exhibit 3 – Qualified Supplier List (QSL)

See Exhibit 4 – Volume Discounts

Notes from Meeting with Supply Chain Analyst

The analyst team is responsible for pulling together information to produce metrics, reports and tools that provide insights into TechnipFMC's supply base. One of these tools is **Supplier Performance Index (SPI) calculator**. This tool is used to risk adjust pricing to factor in a supplier's recent performance. In addition to providing you with the SPI formula, the Global Supply Chain Analyst has also pulled the relevant performance data for the list of **qualified suppliers you received** from the category teams (see Exhibit 3 – Qualified Supplier List).

See Exhibit 5 – SPI Calculator

Notes from Meeting with Supplier Quality

Supplier development is an important tool for improving the performance of underperforming suppliers. Deploying an **Supplier Development Engineer (SDE)** will reduce the supplier's SPI score by .25. The cost of deploying an SDE is \$100,000. Due to diminishing returns, **deployment of two SD Engineers will reduce the supplier's SPI by .35, while three SD Engineers will reduce the SPI by .4**. There are currently 8 SD Engineer's available for this project.

- Due to the high Price of Non Conformance (PONC) associated with this project an **SPI score greater than 1.25 is unacceptable**.

Notes from Meeting with Engineering

The engineering team is currently making changes to two of the core tree components. These engineering changes are driven by new customer requirements. The Lead Project Engineer has warned you that **the following components are still being designed and are not, at this time, available for procurement**.

- Wing Blocks Expected Release: Feb 15th
- Gate Valve Expected Release: Mar 1st

Notes from Meeting with Procurement Team

The Director of Procurement has consolidated the feedback from the team.

See Exhibit 6 – Bid Information

Notes from Meeting with Logistics

1. Shipments to the plant will be staged at various distribution points within each region. As a result, transit time and cost will be consistent across each region. Note that all Incoterms will be FCA.

See Exhibit 7 – Transit Time and Cost